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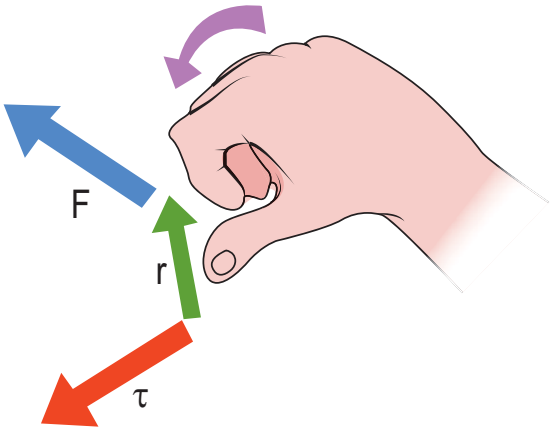


Figure 5.5. Direction of torque using right hand rule

of the paper, then the rotation produced by the torque is anticlockwise. On the other hand, if the direction of the torque is into the paper, then the rotation is clockwise as shown in Figure 5.6.

In many cases, the direction and magnitude of the torque are found separately. For direction, we use the vector rule or right hand rule. For magnitude, we use scalar form as,

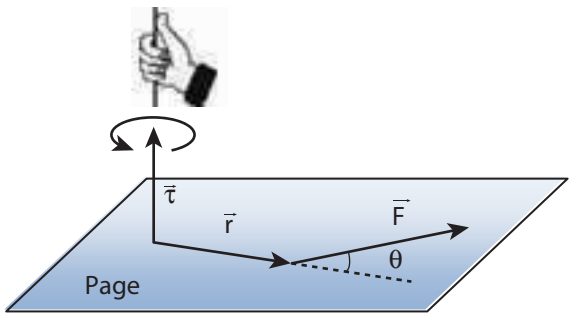
$$\tau = r\ F\ \sin\theta \qquad (5.11)$$

The expression for the magnitude of torque can be written in two different ways by associating $\sin\theta$ either with r or F in the following manner.

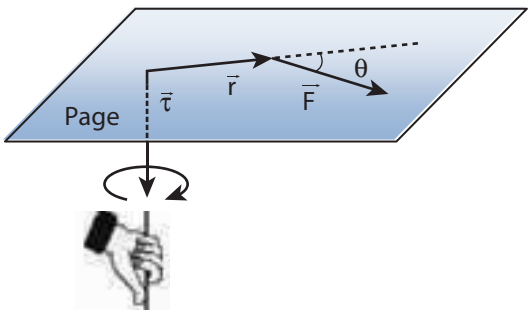
$$\tau = r\left(F\ \sin\theta\right) = r \times \left(F\perp\right) \qquad (5.12)$$

$$\tau = \left(r\ \sin\theta\right)F = \left(r\perp\right) \times F \qquad (5.13)$$

Here, $\left(F\sin\theta\right)$ is the component of $\vec{F}$ perpendicular to $\vec{r}$. Similarly, $\left(r\sin\theta\right)$ is the component of $\vec{r}$ perpendicular to $\vec{F}$. The two cases are shown in Figure 5.7.

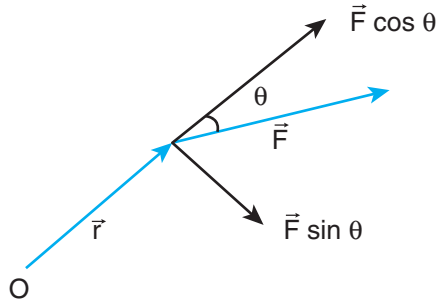


(a) anticlockwise rotation

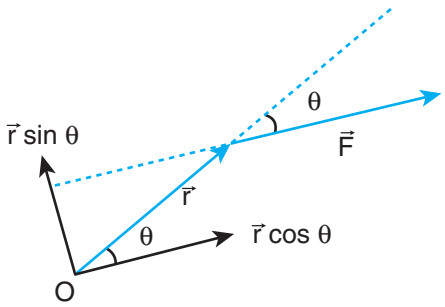


(b) clockwise rotation

Figure 5.6. Direction of torque and the type of rotation



(a) τ = r (F sin θ) = r(F⊥)



(b) τ =(r sin θ) F = (r⊥)F

Figure 5.7. Two ways of calculating the torque.